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(54) **SEAMLESS CIRCULAR KNIT FABRIC, A METHOD OF MAKING SAME AND A FORMING MACHINE USING THIS METHOD, AS WELL AS ARTICLES COMPRISING STRAPS OR BANDS MADE THEREFROM**

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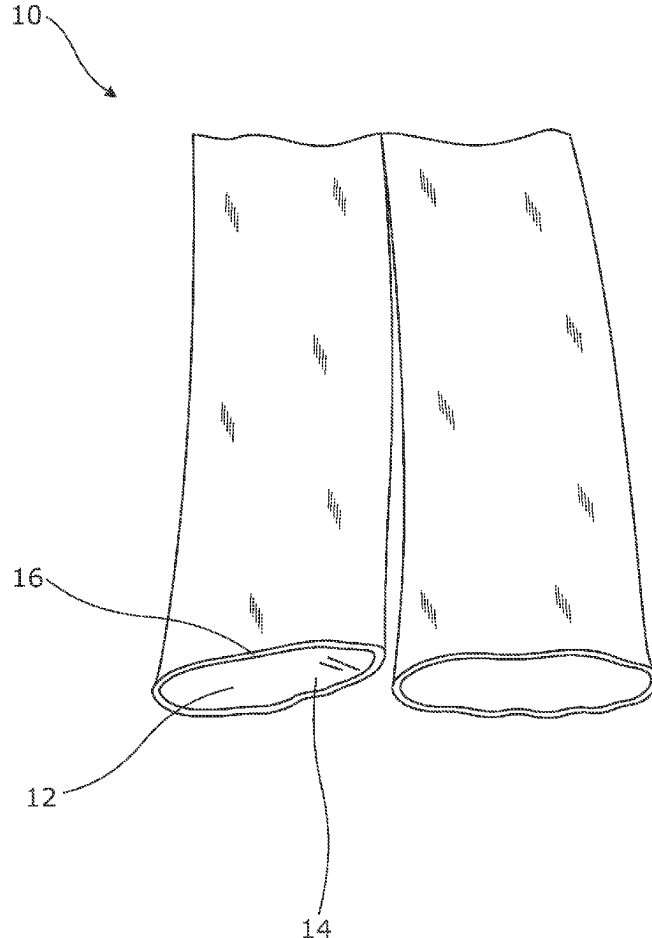
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ABSTRACT

A seamless circular knit fabric (10) comprising an elastic yarn is provided. The fabric (10) has a continuous circular knit through the length thereof. The circular knit fabric (10) further includes a cured cushion filling (12) inside the circular knit fabric (10) and bonded thereto. The method of making same comprises knitting a yarn to produce a continuous length of circular knit fabric (10), and injecting a curable cushion filling in a pre-cured state inside the circular knit fabric (10). This method uses a circular knit fabric forming machine assembly comprising a circular knitting machine and an injector.



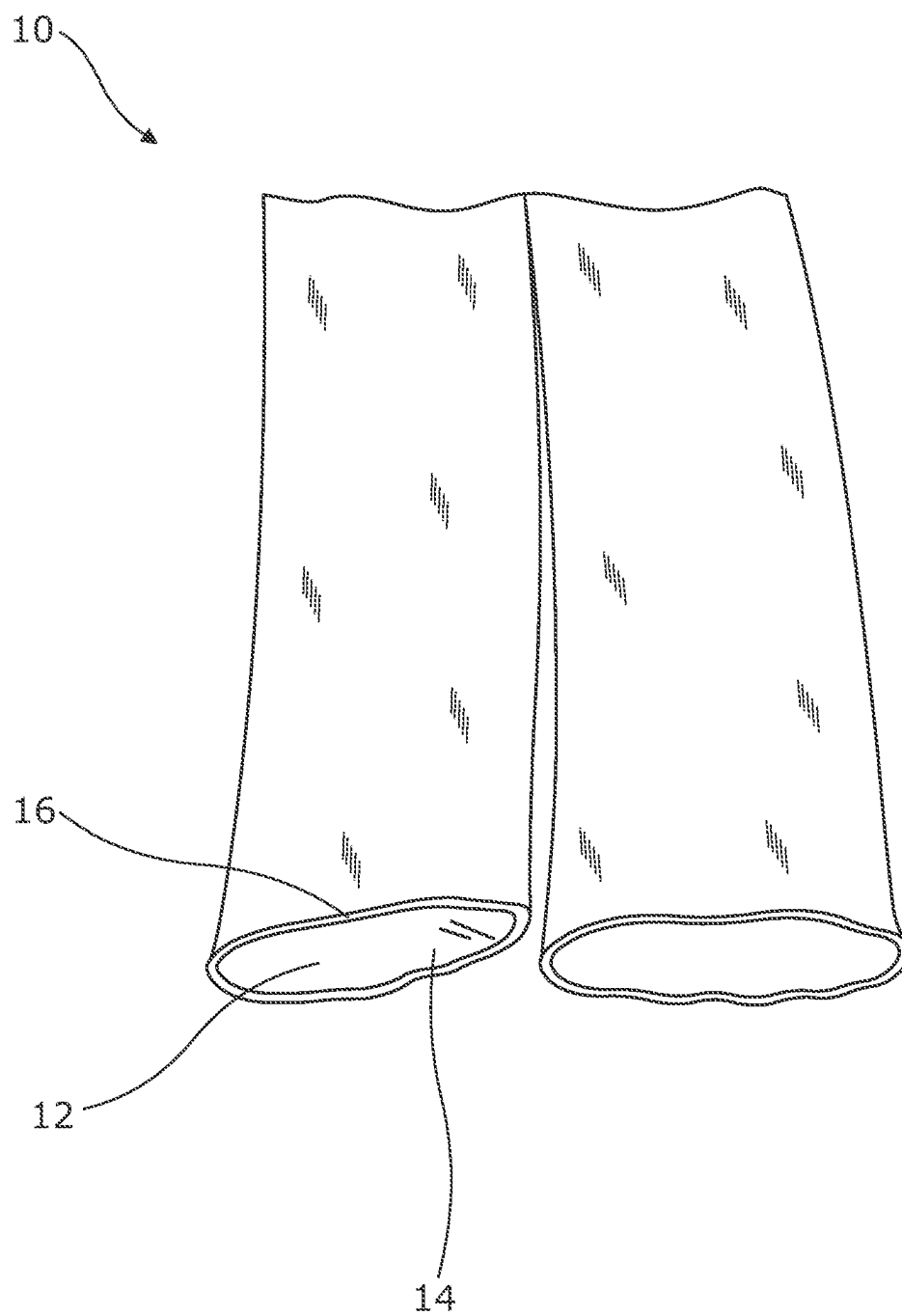


Figure 1a

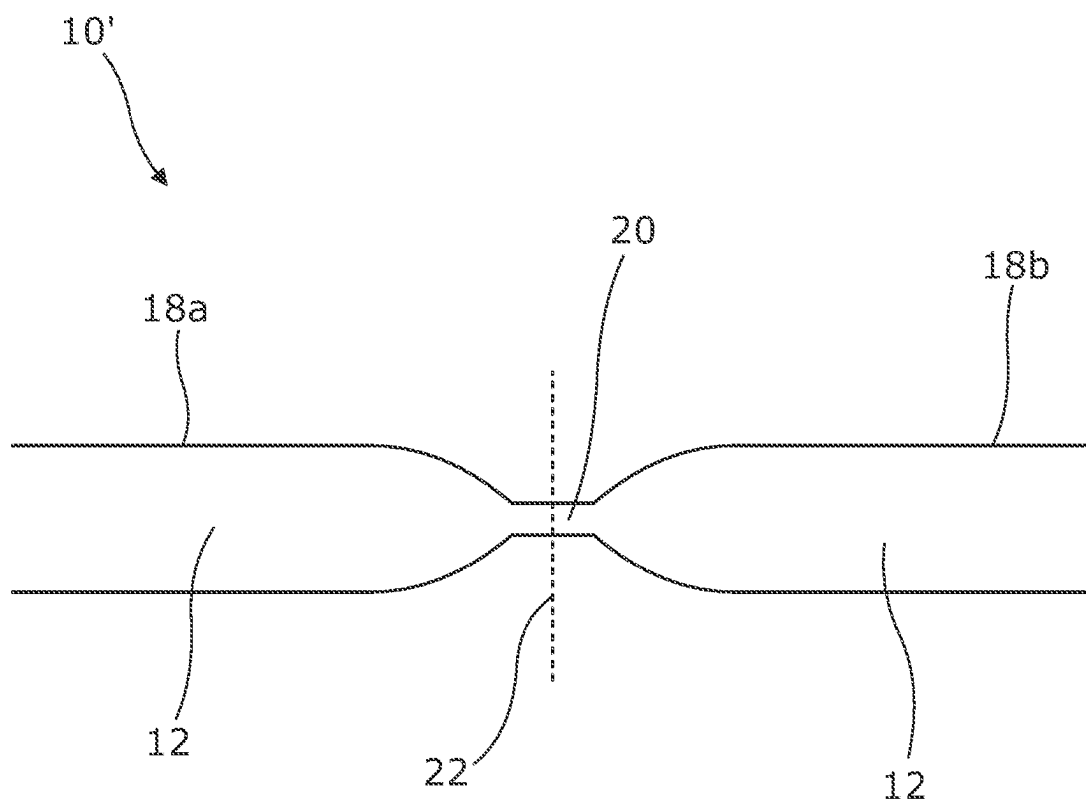


Figure 1b

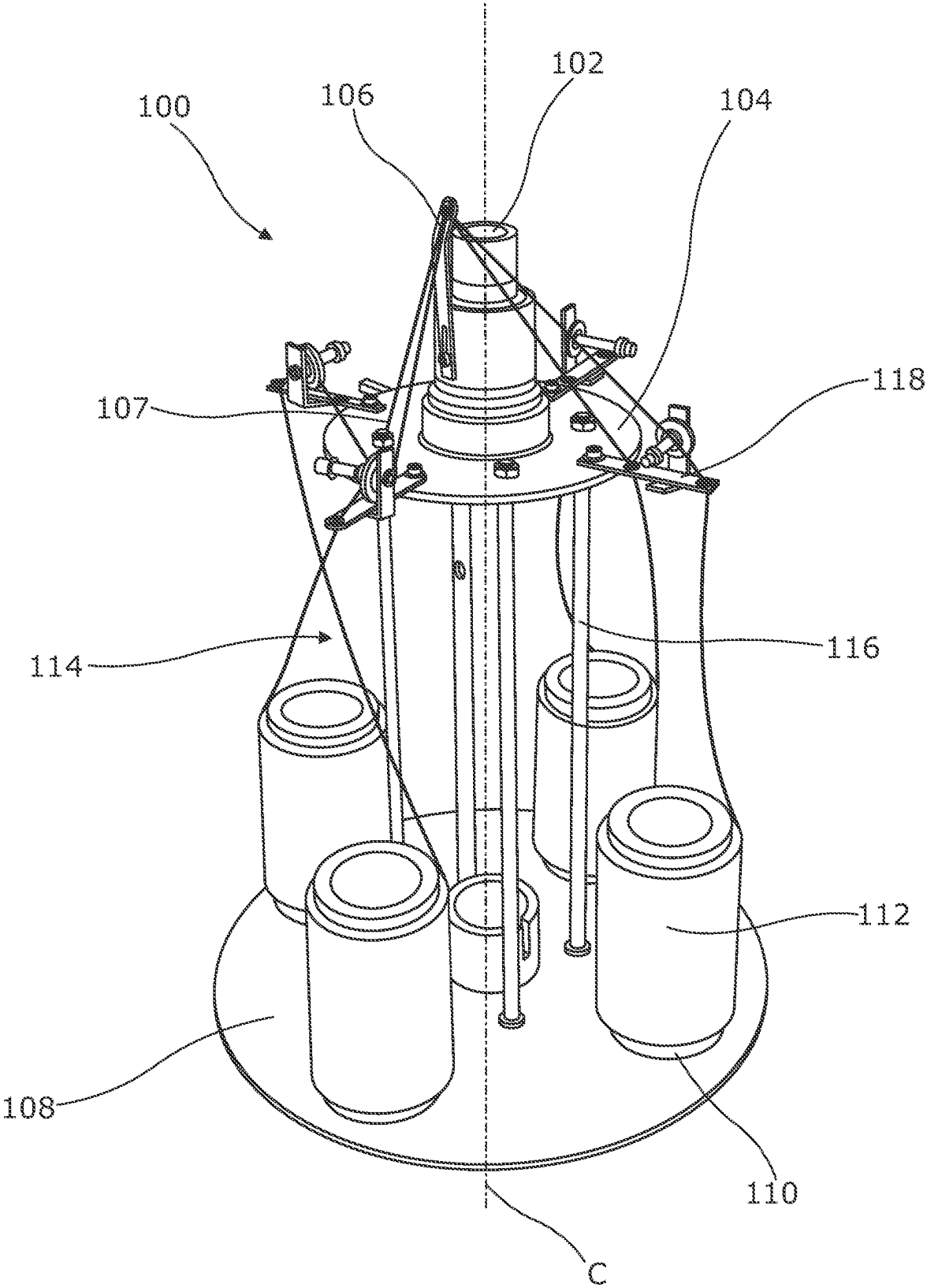


Figure 2a

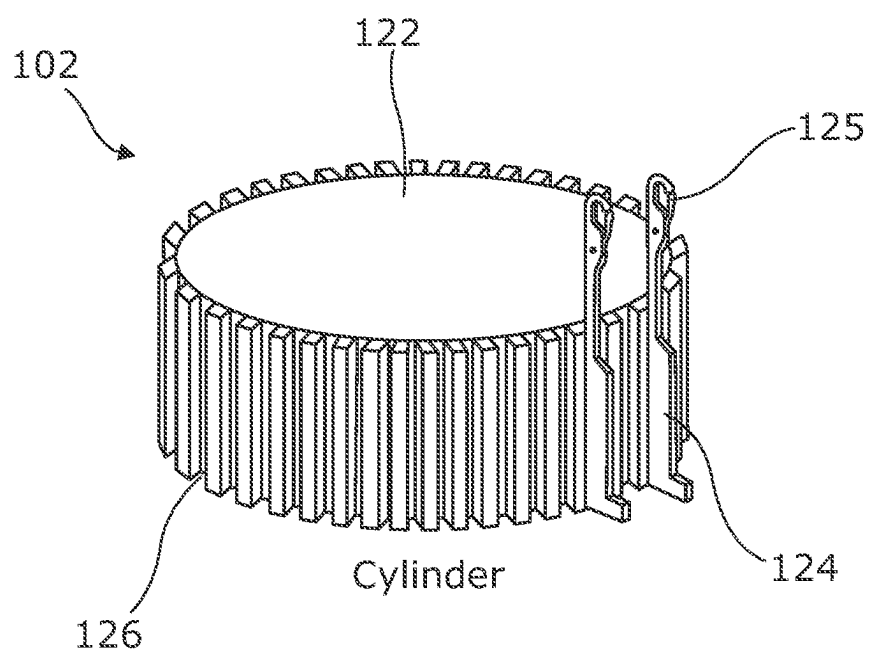


Figure 3a

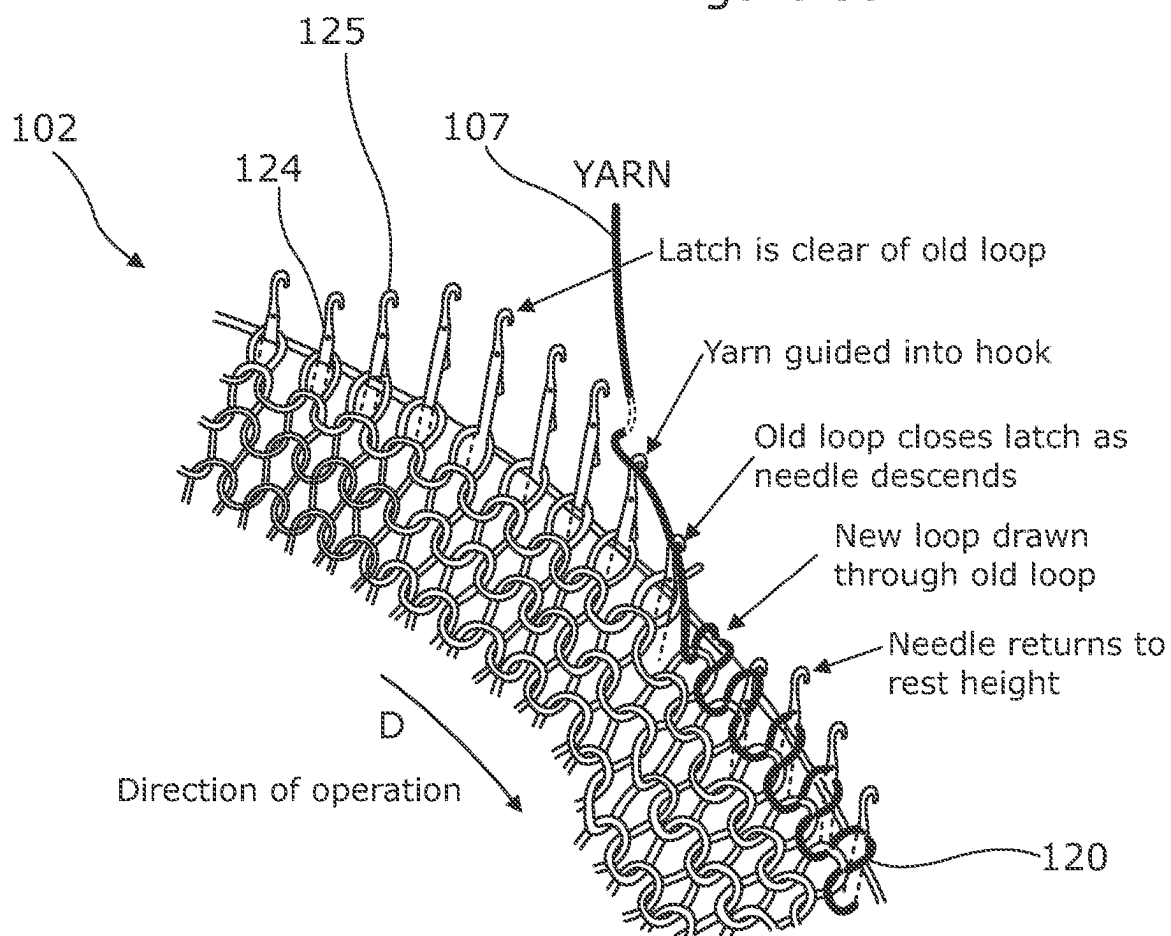


Figure 3b

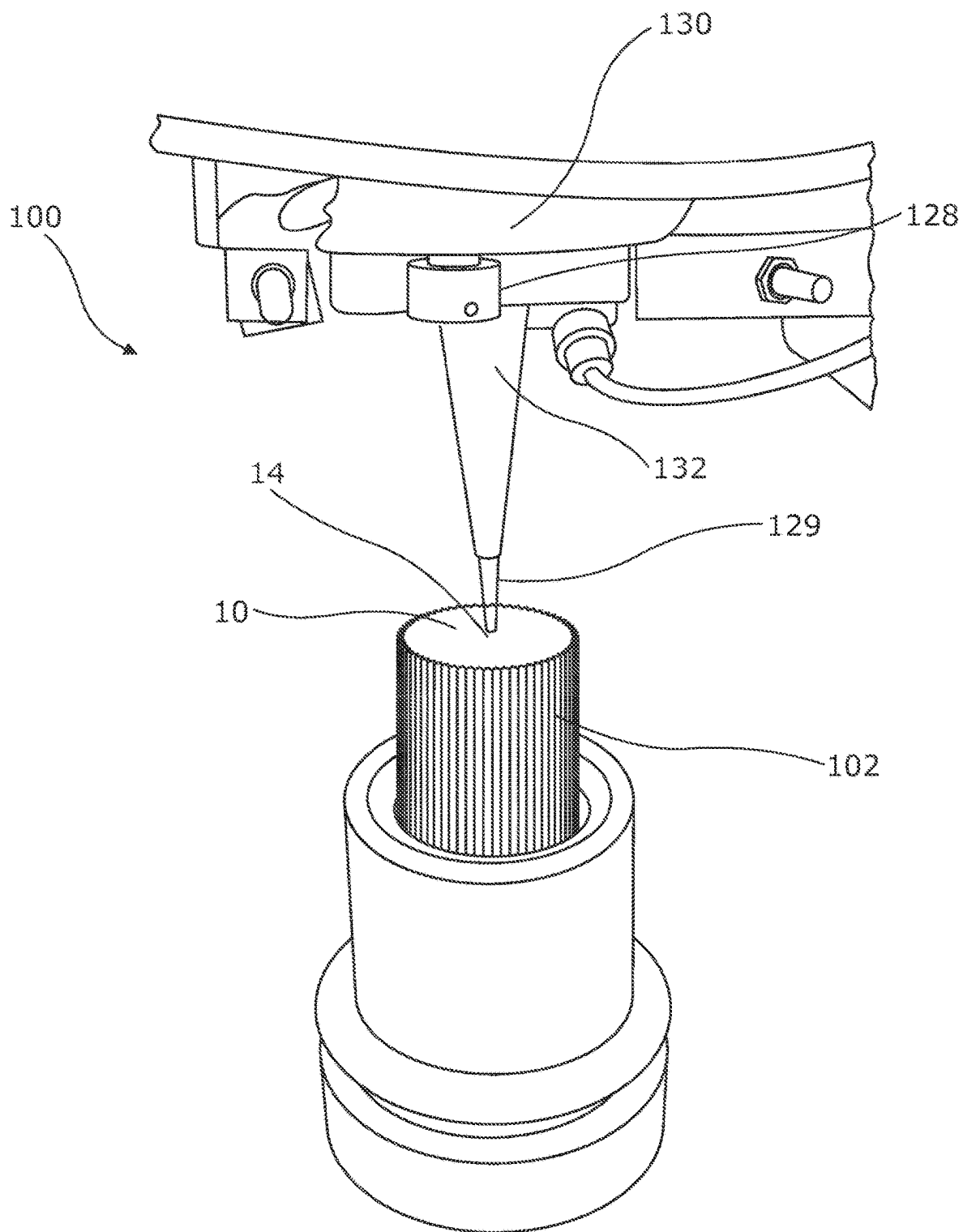


Figure 4

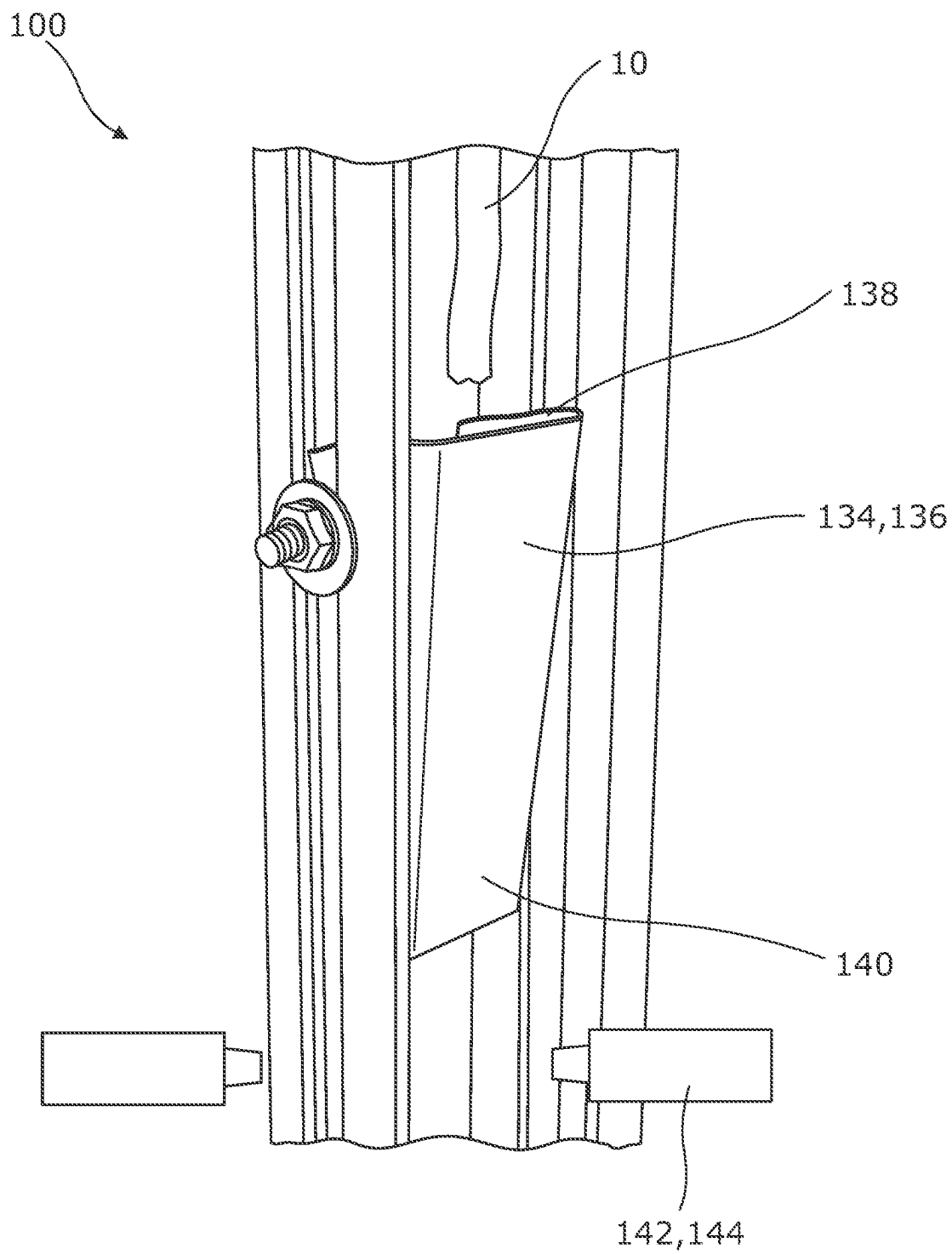


Figure 5

**SEAMLESS CIRCULAR KNIT FABRIC, A
METHOD OF MAKING SAME AND A
FORMING MACHINE USING THIS
METHOD, AS WELL AS ARTICLES
COMPRISING STRAPS OR BANDS MADE
THEREFROM**

[0001] This invention relates to a seamless circular knit fabric, a bra comprising a shoulder strap made from such seamless circular knit fabric, a method of forming a seamless circular knit fabric and a seamless circular knit fabric forming machine assembly.

[0002] Shoulder straps of a bra are typically formed from strips of elastic material which have uncomfortable and conspicuous edge binding material. These materials are relatively hard on a wearer's skin. Traditional shoulder straps tend to dig into the wearer's skin especially as bras are worn by the wearer for the majority of the day. It is also known to use similar strips of material for ear loops which may be used for positioning a face mask onto a wearer's face. Again, the relatively hard ear loop elastic material has a tendency to dig into the wearer's face which is not only uncomfortable but may also leave marks on the skin. The same can be said for the underband of a bra (the part that extends around the back of the wearer) and stretchable waistbands, e.g. for briefs, pants, shorts or trousers/leggings.

[0003] According to a first aspect of the invention there is provided a seamless circular knit fabric comprising an elastic yarn, the fabric having a continuous circular knit through the length thereof, the circular knit fabric further including a cured cushion filling inside the circular knit fabric and bonded thereto.

[0004] A seamless circular knit fabric in the context of the invention is understood to be a seamless tubular fabric. This is different to joining together two pieces of fabric or two edges of the same fabric to produce a tubular fabric, which would not be seamless because of the joint.

[0005] The fabric having a continuous circular knit through its length means that the circular knit pattern flows continuously throughout the length of the fabric. The fabric can be cut to any size, depending on the desired application of the fabric, to create a discrete length of circular knit fabric. Therefore, a user is not restricted by a pre-set length of tubular fabric which circular knitting machines typically produce (e.g. a circular knit sock machine). Moreover, such typical circular knitting machines tend to release a finished single piece of fabric at a pre-set length (e.g. a sock which has its heel and toe knitted), such that the single piece of fabric is not considered to have a continuous circular knit through the length thereof.

[0006] Meanwhile, the seamless circular knit fabric having a cured cushion filling provides additional comfort to the circular knit fabric which may be used as or as part of a wearable garment or accessory. The cured cushion filling being inside the continuous fabric and bonded thereto means that the filling is integral to the fabric and thus provides robust and consistent comfort throughout the length of the fabric. In particular, the filling is added during the manufacturing process of the circular knit fabric which is why it is present in the continuous length of circular knit fabric and bonded thereto (as opposed to it being added after as a separate component to the fabric). The cushion filling being "cured" is understood to mean that it is made from a material which was previously in a non-cured (e.g. soft or liquid) state prior to it curing. As explained in more detail later on,

this means that the cushion filling can be added to the circular knit fabric and cured during the knitting process.

[0007] Preferably, the cured cushion filling is a thermoplastic or thermosetting polymer. Optionally, the cured cushion filling is a two-part silicone gel.

[0008] Any such arrangement allows for the cushion filling to be in a pre-cured state during manufacture of the circular knit fabric which allows it to flow into and fill the hollow cavity of the circular knit fabric as it is being made. It also permits the shape of the fabric (with the pre-cured cushion filling therein) to be moulded to a particular shape, as described in more detail later on.

[0009] The seamless circular knit fabric may have a cut end to define a discrete length of circular knit fabric.

[0010] By the fabric having a cut end it is understood that the fabric as a whole is cut, i.e. the yarn through the circumference of the fabric is cut, so as to create a discrete length of continuous circular knit fabric which can be any desired length. This differs from a single piece of fabric that is released from a machine by cutting the yarn fed into the machine. Such single pieces of fabric would have the knitting finished by the machine at the pre-set length before being released, such that the single piece of fabric is not considered to have a cut end.

[0011] Optionally the seamless circular knit fabric has more than one filled portion, the filled portion being filled with the cured cushion filling, wherein each filled portion is separated by a non-filled portion, the non-filled portion being absent of the cured cushion filling.

[0012] Such an arrangement of neighbouring filled and non-filled portions of the circular knit fabric allows for ease of cutting the continuous length of circular knit fabric at the non-filled portions to form discrete lengths of fabric. Moreover, such cutting would result in the discrete lengths of fabric having non-filled ends which makes it easier to attach the circular knit fabric to another fabric or garment.

[0013] Preferably, the seamless circular knit fabric may have a minimum width of 2 mm, optionally 6 mm, optionally 8 mm. The seamless circular knit fabric may have a maximum width of 20 mm, optionally 30 mm, optionally 50 mm.

[0014] Such minimum widths permit the circular knit fabric to be suitable for use as an ear loop (e.g. minimum of 2 mm) or a bra shoulder strap (e.g. minimum of 6 or 8 mm).

[0015] According to a second aspect of the invention, there is provided a bra comprising a body portion for positioning around a torso of a wearer, and a shoulder strap secured at each end to the body portion, the shoulder strap being made from the seamless circular knit fabric as defined hereinabove.

[0016] According to a third aspect of the invention, there is provided a bra comprising an underband portion for positioning around a torso of a wearer, the underband portion being made from the seamless circular knit fabric as defined hereinabove.

[0017] According to a fourth aspect of the invention there is provided a bottom garment comprising a stretchable waistband for positioning around a waist of a wearer and a bottom portion extending from the waistband, the waistband being made from the seamless circular knit fabric as defined hereinabove.

[0018] It will be understood that the bottom portion may extend from the waistband down the full length of the wearer's leg (e.g. trousers or leggings) or only a portion of

the length of the wearer's leg (e.g. shorts, boxer shorts, skirt, cropped trousers or leggings) or only as far as the wearer's buttocks (e.g. pants, briefs).

[0019] According to a fifth aspect of the invention there is provided a face mask comprising a face covering portion for positioning over a wearer's nose and mouth, and a head strap secured at each end to the face covering portion, the head strap being made from the seamless circular knit fabric as defined hereinabove.

[0020] It will be understood that the "head strap" may be a strap that goes all the way around the wearer's head or may loop around the wearer's ear (i.e. two ear loops).

[0021] Using the seamless circular knit fabric as described hereinabove as a bra shoulder strap, bra underband, waistband or a face mask head strap provides optimal comfort to the wearer firstly because there are no joints in the strap/band that may irritate or dig into the wearer's skin and secondly because the cushion filler provides additional softness and comfort to the wearer.

[0022] According to a sixth aspect of the invention there is provided a method of forming a seamless circular knit fabric comprising:

[0023] feeding an elastic yarn into a circular knit machine head,

[0024] knitting the yarn by bending the yarn into loops and intermeshing yarn loops in a horizontal direction in a seamless manner to produce a continuous length of circular knit fabric,

[0025] injecting a curable cushion filling in a pre-cured state inside the circular knit fabric as the fabric is being knitted.

[0026] The advantages set out above in relation to the features of the first aspect of the invention apply mutatis mutandis to the fourth aspect of the invention.

[0027] It will be understood that injecting the curable cushion filling in a pre-cured state inside the circular knit fabric as the fabric is being knitted means that the pre-cured filling is injected into the hollow cavity formed by the circular knit fabric.

[0028] Preferably, the step of injecting a curable cushion filling includes injecting a thermoplastic or thermosetting polymer.

[0029] Optionally the step of injecting a curable cushion filling includes injecting a two-part silicone gel.

[0030] In an embodiment, the method includes curing the cushion filling as the continuous length of circular knit fabric is being made.

[0031] By curing the cushion filling as the continuous length of circular knit fabric is being made, the cured cushion filling is formed during the same knitting process as that forms the circular knitted fabric, thus reducing the processing steps required. Moreover, the curing step results in a cured cushion filling that forms part of (i.e. bonded to) the circular knit fabric, thus providing a continuous and consistent cushioning throughout the length of the circular knit fabric.

[0032] Optionally the method includes sequentially filling and ceasing to fill the circular knit fabric with the curable cushion filling in a pre-cured state to form neighbouring filled and non-filled portions of the circular knit fabric.

[0033] The method may include passing the continuous length of circular knit fabric with curable cushion filling through an adjustment mechanism to adjust the shape of the fabric.

[0034] Such a step makes use of the mouldable nature of the pre-cured cushion filling to produce a circular knit fabric with a desired shape.

[0035] The method may include feeding the yarn from a yarn platform upwards towards the circular knit machine head, and producing the continuous length of circular knit fabric in a space created between the circular knit head and the yarn platform.

[0036] Such an arrangement provides space for the circular knit fabric to be produced in a continuous manner without restricting the length of the fabric being produced. Moreover, creating a space between the yarn platform and the circular knit machine head allows for additional processing steps to readily take place during the knitting process because such additional mechanisms (e.g. an adjustment mechanism) can operate in that space between the yarn platform and the circular knit machine head. In addition to the foregoing, it allows the injection of the pre-cured cushion filling to happen directly at the machine head without obstruction from other parts of the machinery. This is important to prevent premature curing of the cushion filling which may otherwise occur of the cushion filling has a further distance to travel to the machine head.

[0037] The method may include pulling the continuous length of circular knit fabric in the direction that the circular knit fabric is being produced.

[0038] Such a pulling step (e.g. performed by a takedown mechanism) helps to dictate the courses per inch (CPI) or the fabric.

[0039] Preferably, the method further includes cutting the continuous length of circular knit fabric to discrete lengths of circular knit fabric.

[0040] Cutting the continuous length of circular knit fabric permits the creation discrete lengths of continuous circular knit fabric which can be any desired length.

[0041] The method may include producing a continuous length of circular knit fabric having a minimum width of 2 mm, optionally 6 mm, optionally 8 mm. The method may include producing a continuous length of circular knit fabric having a maximum width of 20 mm, optionally 30 mm, optionally 50 mm.

[0042] According to a seventh aspect of the invention there is provided a seamless circular knit fabric forming machine assembly comprising:

[0043] a circular knit machine head configured to bend elastic yarn into loops and intermesh the yarn loops in a horizontal direction in a seamless manner to produce a continuous length of circular knit fabric,

[0044] a yarn feeder configured to feed yarn into the circular knit machine head,

[0045] an injector configured to inject a curable cushion filling in a pre-cured state inside the circular knit fabric as it is being knitted.

[0046] The advantages set out above in relation to the features of the first and fourth aspects of the invention apply mutatis mutandis to the fifth aspect of the invention.

[0047] The injector may be configured to inject a thermoplastic or thermosetting polymer.

[0048] The injector may be configured to inject a two-part silicone gel.

[0049] Optionally the injector is configured to sequentially fill and cease to fill the circular knit fabric with the curable cushion filling in a pre-cured state to form neighbouring filled and non-filled portions of the circular knit fabric.

[0050] Optionally, the machine assembly includes a curing mechanism configured to apply a curing effect to the continuous circular knit fabric to achieve curing of the pre-cured cushion filling.

[0051] Preferably, the machine assembly includes an adjustment mechanism configured to receive the continuous circular knit fabric with pre-cured cushion filling and to adjust the shape of the fabric as a result of the fabric passing through the adjustment mechanism.

[0052] The machine assembly may include yarn platform wherefrom the yarn is fed upwards towards the circular knit machine head, wherein the continuous circular knit fabric is produced in a space created between the circular knit head and the yarn platform.

[0053] The machine assembly may include a takedown mechanism configured to pull the continuous length of circular knit fabric in the direction that the circular knit fabric is being produced.

[0054] Optionally the machine assembly includes a cutting mechanism configured to cut the continuous length of circular knit fabric into discrete lengths of fabric.

[0055] The circular knit machine head may be configured to produce a continuous length of circular knit fabric having a minimum width of 2 mm, optionally 6 mm, optionally 8 mm.

[0056] The circular knit machine head may be configured to produce a continuous length of circular knit fabric having a maximum width of 20 mm, optionally 30 mm, optionally 50 mm.

[0057] A preferred embodiment of the invention will now be described, by way of a non-limiting example, with reference to the accompanying drawings in which:

[0058] FIG. 1a shows a seamless circular knit fabric according to a first embodiment of the invention;

[0059] FIG. 1b shows the seamless circular knit fabric according to a second embodiment of the invention;

[0060] FIGS. 2a and 2b show a seamless circular knit fabric forming machine assembly according to a third embodiment of the invention;

[0061] FIG. 3a shows the needle bed of the machine head from the machine assembly shown in FIGS. 2a and 2b;

[0062] FIG. 3b shows the direction of operation of the needle bed shown in FIG. 3a;

[0063] FIG. 4 shows an injector portion of the machine assembly shown in FIGS. 2a and 2b; and

[0064] FIG. 5 shows an adjustment and curing portions of the machine assembly shown in FIGS. 2a and 2b.

[0065] A seamless circular (or tubular, as it may also be referred to) knit fabric according to an embodiment of the invention is shown in FIGS. 1a and 1b and is designated by reference numeral 10.

[0066] The circular knit fabric 10 has a degree of stretch due to the inclusion of an elastic yarn. The circular knit fabric 10 has a continuous circular knit through the length thereof, which is discussed in more detail later.

[0067] The circular knit fabric 10 further includes a cured cushion filling 12 inside the circular knit fabric 10. That is to say that the cured cushion filling 12 fills what would be the hollow cavity 14 of the circular knit fabric 10. The cured cushion filling 12 is bonded to the circular knit fabric 10.

[0068] In the embodiment shown, the cured cushion filling is a thermoplastic elastomer (TPE) and more specifically may be thermoplastic polyurethane (TPU).

[0069] In other embodiments, the cured cushion filling 12 may instead be any suitable thermoplastic or thermosetting polymer. The cured cushion may be a two-part silicone gel. The silicone gel may be of a type known as RTV 2K GEL TP 3841™ or “SILOPREN” gel TP 3904 as sold by GE Bayer Silicones GmbH & Co. KG. The cured cushion filling may instead be a curable foam or a polymer gel. The cured cushion filling may instead be a bio-based polymer, biodegradable synthetic polymer, or naturally occurring biodegradable/non-degradable polymer.

[0070] The circular knit fabric 10 has a cut end 16 which defines a discrete length of circular knit fabric. The circular knit fabric 10 leaves the knitting machine as a continuous length of circular knit fabric 10, which can then be cut into desired lengths. This is opposed to the circular knit fabric being finished as a pre-set length single piece of fabric.

[0071] In this embodiment, the cured cushion filling 12 fills the length of circular knit fabric 10 in a continuous manner such that even when it is cut, the cured cushion filling 12 fills right up to the cut end 16.

[0072] In the embodiment shown, the width of the circular knit fabric 10 is approximately 8 mm. The width may instead be as narrow as 2 mm (e.g. ear loop) or may be as wide as 50 mm (e.g. waistband). The continuous length of circular knit fabric 10 that is produced may vary but may be at least 40 mm. The output table below indicates exemplary widths of the circular knit fabric 10 and an intended end use of a fabric having such a width. The table also outlines an example of the output per hour and speed of the circular knit machine that may be used (which is discussed in more detail later on).

Output Table 1

Circular knit fabric width	Output per hour	Speed of machine	End use
2 mm-6 mm		1000 RPM	Ear loop
8 mm-18 mm	Ave. 9M Per one machine head	700 RPM	Bra Strap
20 mm-30 mm	Ave. 9M Per one machine head	600 RPM	Bra Strap

[0073] The finished circular knit fabric width may have the following tolerances:

[0074] 2 mm-6 mm accurate

[0075] 8 mm to 18 mm: +/-0.5 mm

[0076] 20 mm to 30 mm: +/-1 mm

[0077] The circular knit fabric may be formed from a single elastic yarn. The elastic yarn may be Nylon. For example, the elastic yarn may be 2/78/24/0 Nylon 66 (“2/78/24/0” being a common way to represent characteristics of a yarn, i.e. representing 2 ply, 78 denier, 24 filaments, 0 twists per meter (TPM)).

[0078] The circular knit fabric may be formed from more than one yarn. Both or all yarns may be the same type of yarn or they may instead be different yarns. The yarns may have different properties. For example, the yarn may be any single cover yarn.

[0079] A circular knit fabric 10' according to a second embodiment of the invention is shown in FIG. 1b. The circular knit fabric 10' of the second embodiment shared identical features to the circular knit fabric 10 of the first embodiment and like features are indicated by the same reference numerals.

[0080] As shown in FIG. 1*b*, the circular knit fabric 10' in this embodiment includes first and second filled portions 18*a*, 18*b* which are filled with the cured cushion filling 12. The first and second filled portions 18*a*, 18*b* are separated by a non-filled portion 20 which is absent of any cured cushion filling 12.

[0081] Although only two filled portions 18*a*, 18*b* are shown in FIG. 1*b*, it will be understood that there may be any number of filled portions 18*a*, 18*b* separated by non-filled portions 20.

[0082] The non-filled portion 20 is considerably shorter in length to the filled portions 18*a*, 18*b*. In other embodiments, the lengths of the filled and non-filled portions 18*a*, 18*b*, 20 may be similar or the same as each other. In further embodiments, the lengths of each portion 18*a*, 18*b*, 20 may differ (i.e. the first filled portion 18*a* may have a different length to the second filled portion 18*b*).

[0083] Moreover, each of the first and second filled portions 18*a*, 18*b* may be filled with different types of cured cushion filling 12.

[0084] The non-filled portion 20 presents a section which can be easily cut (i.e. at line 22) to form discrete lengths of circular knit fabric 10', each of which having a non-filled cut end.

[0085] A seamless circular knit fabric forming machine assembly is shown in FIGS. 2*a* to 5 and is designated by reference numeral 100.

[0086] The machine assembly 100 has a circular knit machine head 102 secured to a knit head platform 104. The machine assembly 100 also includes a yarn feeder 106 configured to feed yarn 107 into the circular knit head machine 102.

[0087] The machine assembly 100 further includes a yarn platform 108 from which the yarn 107 is supplied. In particular, the yarn platform 108 includes bobbin holders 110 (in the form of pegs 110 in this example) onto which yarn bobbins 112 locate in use. In the example shown, there are four pegs 110 and therefore four yarn bobbins 112 are positioned on the yarn platform 108. The yarn bobbins 112 are stood in an upright manner relative to the yarn platform 108. As can be seen, the yarn pegs 110 are positioned around the centre of the yarn platform 108.

[0088] The knit head platform 104 and the yarn platform 108 are spaced from one another such that a gap 114 is created between the circular knit machine head 102 and the yarn platform 108. Moreover, the knit head platform 104 is positioned higher than the yarn platform 108. In other words, there is a space 114 between where the yarn 107 is being supplied and where the yarn 107 is being fed into the machine, and the yarn 107 is being fed to the feeder 106 in an upwards manner.

[0089] In the embodiment shown, the knit head platform 104 is positioned directly above the yarn platform 108. Moreover, the knit head platform 104 and the yarn platform 108 are centrally aligned with one another. Both platforms 108, 104 are circular platforms which share a central axis C. In other embodiments, the platforms 108, 104 may take any shape and/or may be out of line with one another.

[0090] The space 114, i.e. the distance between the yarn platform 108 and the knit head platform 104 may be around 40 cm.

[0091] The machine assembly 100 includes a support member 116 which secures the platforms 104, 108 to one another and spatially positions them relative to one another. In the embodiment shown, there are four support members

116 that are equally spaced around the centre of each platform 104, 108. The support member 116 may take any suitable form.

[0092] The machine assembly 100 includes yarn guides 118 which guide the yarn 107 from the yarn platform 108 towards the yarn feeder 106. The yarn guides 116 help to maintain a desired tension in the yarn 107.

[0093] In the embodiment shown, there are four yarn guides 118 which are secured to the knit head platform 104. Each yarn guide 116 is equally spaced from one another around the centre of the knit head platform 104. Each yarn guide 118 is positioned in line with a peg 110 and receives yarn 107 from the yarn bobbin 112 which is situated on the peg 110 directly below the respective yarn guide 118. There is a single yarn feeder 106 and the yarn guides 118 feed yarn 107 into the single yarn feeder 106.

[0094] In the embodiment shown, the yarn 107 from two of the yarn bobbins 112 is fed directly from the yarn bobbin 112 through the respective yarn guide 118 to the yarn feeder 106. Meanwhile the yarn 107 from the remaining two of the yarn bobbins 112 is fed from the yarn bobbin 112 through the respective yarn guide 118 and then via a neighbouring yarn guide 118 to the yarn feeder 106. This is shown more clearly in FIG. 2*b* in which the direct yarn path Y_D from two of the yarn bobbins 112*a* is shown and the indirect yarn path Y_I of the remaining two yarn bobbins 112*b* is shown.

[0095] Turning to FIGS. 3*a* and 3*b*, the circular knit machine head 102 is configured to bend the yarn 107 into loops 120 and intermesh the yarn loops 120 in a horizontal direction D in a seamless manner so as to produce a continuous length of circular knit fabric.

[0096] The circular knit machine head 102 includes a needle bed 122 which holds a plurality of needles 124 in a cylindrical manner (as shown in FIG. 3*a*—only two needles shown for clarity purposes). The needles 124 have a hook 125 at one end. The needles 124 are held at specific distances relative to one another, which is determined by the needle bed 122. The needle bed 122 also guides the needles 124 during the knitting process. To achieve this control, the needle bed 122 includes needle slots 126 (also referred to as needle tracks) in which a respective needle 124 is located and the slots 126 act to guide each needle 124 during knitting.

[0097] As can be seen in FIG. 3*b*, the needles 124 move up and down in turn in a horizontal (when looking directly at the machine head 102) direction of travel D, with each needle 124 moving up to catch (via its hook 125) the yarn 107 being fed into the circular knit machine head 102 and then pulling that yarn 107 down and through a yarn loop 120 already created in the fabric.

[0098] The number of slots 126 in the circular knit machine head 102 determines the knitting gauge. The gauge may be 12 to 16.

[0099] The diameter of the circular knit machine head 102 influences the width of the circular knit fabric 10 that is produced. The type of yarn 107 that is used is also a factor in the final width of the fabric 10.

[0100] Turning to FIG. 4, the machine assembly 100 further includes an injector 128 which is configured to inject a curable cushion filling 129 in a pre-cured state inside the circular knit fabric 10 as it is being knitted.

[0101] The injector 128 includes an extrusion machine 130 and a nozzle 132. In this embodiment, the extrusion machine 130 extrudes TPE and the nozzle 132 is an inkjet nozzle which inserts TPE in a pre-cured state into the circular knit fabric 10.

[0102] As can be seen, the nozzle 132 points into the hollow cavity 14 of the circular knit fabric 10 at the end where the fabric 10 is being formed.

[0103] In some embodiments, the injector 128 may be configured to sequentially fill and cease to fill the circular knit fabric with the curable cushion filling 129 in a pre-cured state to form neighbouring filled and non-filled portions of the circular knit fabric. Such filled and non-filled portions are described in more detail above in relation to FIG. 1b.

[0104] Turning to FIG. 5, the machine assembly 100 further includes an adjustment mechanism 134. The adjustment mechanism 134 shown in this embodiment is a tapered cavity 136 which receives, at the wider end 138 thereof, the continuous circular knit fabric 10 with the pre-cured cushion filling inside. The fabric 10 passes through the narrower end 140 of the tapered cavity 136 which moulds the shape of the pre-cured cushion filling and therefore the shape of the circular knit fabric 10. For example, the tapered cavity 136 may be an oblong shape at its narrow end 140 and therefore squash the fabric 10 to a flat, thin shape. Other shapes that the adjustment mechanism 134 may produce include oval, circular, rectangular.

[0105] As also shown in FIG. 5, the machine assembly 100 further includes a curing mechanism 142. The curing mechanism 142 is located directly below the adjustment mechanism 134 so as to cure the cushion filling after it has been moulded to a desired shape by the adjustment mechanism 142, therefore setting the desired shape of the circular knit fabric 10.

[0106] In this embodiment, the curing mechanism 142 is an air cooler 144 which blows cool air over the circular knit fabric 10 to cure the TPE cushion filling. In other embodiments, the curing can be done by cool or hot air, water or by adding another material which initiates curing. Moreover, the curing mechanism 142 may be a combination of such curing methods.

[0107] Although not shown in this embodiment, the machine assembly 100 may also include a takedown mechanism which is configured to pull the circular knit fabric 10 down in the direction of travel (i.e. in the direction it is being produced). The machine assembly 100 may also include a cutting mechanism to cut the continuous length of circular knit fabric to desired discrete lengths.

[0108] Exemplary raw materials and machine settings to produce a circular knit fabric 10 of the invention are outlined below.

Example 1

[0109] Intended use: Bra strap

Raw Material Information:

Raw materials 1			
Yarn info	Yarn code	No of Yarn	Yarn weight
Yarn 1 2/78/24/0 Nylon 66 semi dull textured set air mingle	3784	2	2.6 g
Yarn 2 Single cover yarn		2	2.1 g

Machine Settings

- [0110] Loom speed: 700 RPM
- [0111] Loom picks: 3.6
- [0112] Unfinished width: 11 mm
- [0113] Number of tape: 10
- [0114] Unfinished stretch: 90 at 3.5 Kg
- [0115] Finished picks (courses): 33
- [0116] Finished weight (1 M): 5.1 g
- [0117] Machine head diameter: 17 mm
- [0118] Number of needles: 34
- [0119] Needle gauge: 16
- [0120] Noodle size: 16

1. A seamless circular knit fabric comprising an elastic yarn, the fabric having a continuous circular knit through a length thereof, the circular knit fabric further including a cured cushion filling that fills an inside of a cavity of the circular knit fabric and bonded thereto.

2. The seamless circular knit fabric according to claim 1 wherein the cured cushion filling is a thermoplastic or thermosetting polymer.

3. The seamless circular knit fabric according to claim 1 wherein the cured cushion filling is a two-part silicone gel.

4. The seamless circular knit fabric according to claim 1 having a cut end to define a discrete length of circular knit fabric.

5. The seamless circular knit fabric according to claim 1 having more than one filled portion, the filled portion being filled with the cured cushion filling, wherein each filled portion is separated by a non-filled portion, the non-filled portion being absent of the cured cushion filling.

6. The seamless circular knit fabric according to claim 1 having a minimum width of 2 mm, optionally 6 mm, optionally 8 mm.

7. The seamless circular knit fabric according to claim 1 having a maximum width of 20 mm, optionally 30 mm, optionally 50 mm.

8. The seamless circular knit fabric according to claim 1, wherein the fabric forms at least one of:

a shoulder strap secured at each end to a body portion, the body portion for positioning around a torso of a wearer in use;

an underband portion of a bra, the underband portion for positioning around a torso of a the wearer in use;

a stretchable waistband of a bottom garment, the stretchable waistband for positioning around a waist of the wearer; or

a head strap secured at each end to a face covering portion of a face mask.

9. (canceled)

10. (canceled)

11. (canceled)

12. The method according to claim 34 wherein the step of inserting the curable cushion filling includes:

injecting the curable cushion filling in a pre-cured state inside the cavity of the circular knit fabric such that the cavity is filled with the curable cushion filling, wherein the injecting is being carried out as the fabric is being knitted.

13. (canceled)

14. (canceled)

15. The method according to claim 34, including curing the cushion filling as the continuous length of circular knit fabric is being made.

16. (canceled)

17. The method according to claim 34 including passing the continuous length of circular knit fabric with the curable cushion filling through an adjustment mechanism to adjust a shape of the fabric.

18. The method according to claim 34 including feeding the yarn from a yarn platform upwards towards the circular knit machine head, and producing the continuous length of circular knit fabric in a space created between the circular knit head and the yarn platform.

19. The method according to claim 34 including pulling the continuous length of circular knit fabric in the direction that the circular knit fabric is being produced.

20. The method according to claim 34 further including cutting the continuous length of circular knit fabric to discrete lengths of circular knit fabric.

21. (canceled)

22. (canceled)

23. The seamless circular knit fabric forming machine assembly

according to claim 35 wherein the inserter includes an injector configured to inject the curable cushion filling in a pre-cured state inside the cavity of the circular knit fabric so as to fill the cavity with the curable cushion filling as the circular knit fabric is being knitted.

24. (canceled)

25. (canceled)

26. (canceled)

27. (canceled)

28. The seamless circular knit fabric forming machine assembly according to claim 35, further including an adjustment mechanism configured to receive the continuous circular knit fabric with the pre-cured cushion filling and to adjust a shape of the fabric as a result of the fabric passing through the adjustment mechanism.

29. The seamless circular knit fabric forming machine assembly according to claim 35, including a yarn platform

wherefrom the yarn is fed upwards towards the circular knit machine head, wherein the continuous circular knit fabric is produced in a space created between the circular knit head and the yarn platform.

30. The seamless circular knit fabric forming machine assembly according to claim 35, further including a take-down mechanism configured to pull the continuous length of circular knit fabric in the direction that the circular knit fabric is being produced.

31. (canceled)

32. (canceled)

33. (canceled)

34. A method of forming a seamless circular knit fabric comprising:

feeding an elastic yarn into a circular knit machine head; knitting the yarn by bending the yarn into loops and intermeshing yarn loops in a horizontal direction in a seamless manner to produce a continuous length of circular knit fabric;

inserting a curable cushion filling inside a cavity of the circular knit fabric such that the cavity is filled with the curable cushion filling, wherein the inserting is being carried out as the fabric is being knitted.

35. A seamless circular knit fabric forming machine assembly comprising:

a circular knit machine head configured to bend elastic yarn into loops and intermesh the yarn loops in a horizontal direction in a seamless manner to produce a continuous length of circular knit fabric;

a yarn feeder configured to feed yarn into the circular knit machine head; and

an inserter configured to insert a curable cushion filling inside a cavity of the circular knit fabric so as to fill the cavity with the curable cushion filling as the circular knit fabric is being knitted.

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